

CY 521: Mathematics for Chemist

Course Code	CY 521
Course Name	Mathematics for Chemist
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Linear Algebra** [14 Lectures]

Scalars and Vectors: vector, a summary of vectors and representation of 3-D co-ordinates, Vector Products, Vector Spaces, orthonormal basis set, vector functions, differentiation and integration of vector functions, gradient, divergence and Curl, line integral, Surface integral, volume integrals, Linear independence, basis, curvilinear coordinates, Laplacian, Greens theorem, Stokes theorem, Tensors

Matrix Algebra: Matrix and determinants, Rank, Matrix Algebra, diagonalization, Matrix property, Matrix functions, Eigen values, Eigen vectors,

Applications: Visualization in quantum chemistry, Symmetry and group theory, Spectroscopy

- **Calculus** [14 Lectures]

Functions, Variables, Bessel functions, basic rules of differentiation, maxima and minima, exact and inexact differential, partial differentiation (limits), rules of integration, definite and indefinite integrals, first and second order differential equations, linear homogeneous/inhomogeneous equations, general solution and particular solution, polynomials, Taylor and McLaurin series, Fourier series and Fourier transforms, power series method

Calculus of Several Variables: Functions of More Than One Variable, An Introduction to Partial Derivatives, Differentiability and the Gradient, The Chain Rule, Derivatives of Integrals, The Total Differential, the method of Lagrange Multipliers, multiple integrals

Applications: Kinetics, Quantum Chemistry, thermodynamics

Special Functions: integral functions, interpolation, delta function

- **Analytical geometry and Data Analysis** [14 Lectures]

Analytical geometry: Properties of different graphs and plotting of them, understanding of any kind of scientific equation- plotting of them and interpreting the physical significance of the plot. Relevant especially for thermodynamic, kinetic and radioactive study, general application in all fields of chemistry

Data Analysis: Data Plotting, least square fitting, concept of floating points, interpolation, asymptotic analysis, error analysis, random numbers, correlations.

Text Books

1. R.G.Mortimer, **Mathematics for Physical Chemists**, Academic Press, 2005.
2. Charles L. Perrin, **Mathematics for Chemists**, Wiley-Inter science, 1970.
3. Donald A. McQuarrie, **Mathematical Methods for Scientists and Engineers**, University Science Books, 2003.

References

1. E. Steiner, **The Chemical Maths Book**, 2nd Edition, Oxford University Press, 2008.
2. Martin C. R. Cockett and Graham Doggett, **Maths for Chemists**, Volumes 1 and 2, Royal Society of Chemistry, Cambridge Press, 2003.